

Task: Website hosting through S3 bucket and configuration of backend.

Task Details:

- Host the **frontend (HTML/CSS/JS)** on **Amazon S3** using **Static Website Hosting**.
- Deploy the **backend application/API** on a **private EC2 instance** behind an **Application Load Balancer (ALB)**.
- Ensure the backend EC2 instance has **no public access** and is reachable **only via the ALB**.
- Configure **Amazon CloudFront** in front of the S3 bucket to serve the frontend.
- Update the frontend code to call the backend using the **ALB DNS name**
(Example: <http://<ALB-DNS>/api>).

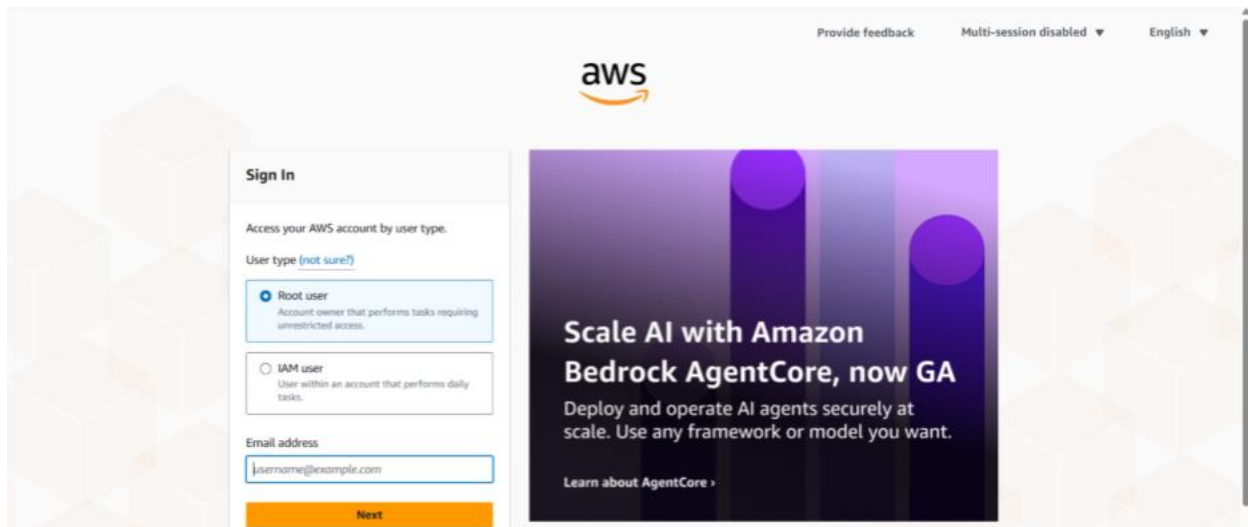
To cross verify your setup ,you can used this command

curl <http://<ALB-DNS>>

Solution:

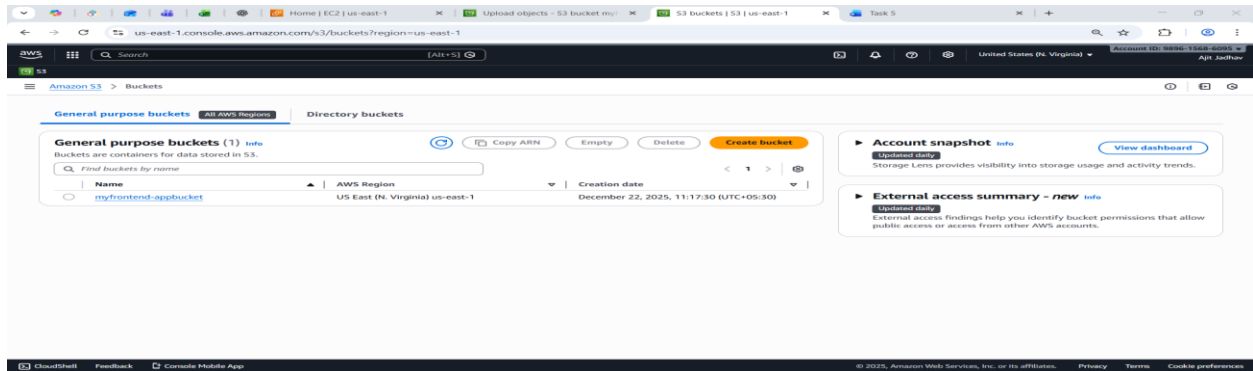
Login to your AWS account with credentials and required verification.

1. (<https://signin.aws.amazon.com/>)
My account – AWS Management Console- Sign in



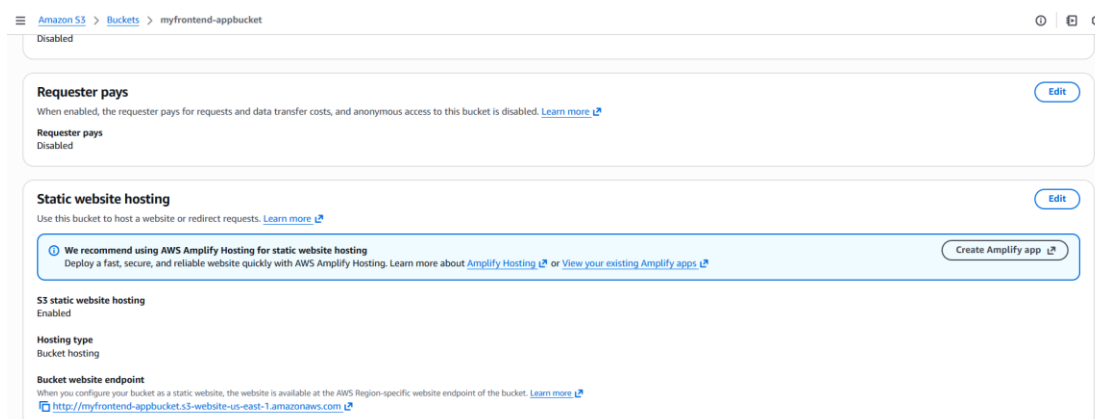
2. Create S3 Bucket
 - Goto S3- create bucket-

- Enter:
- Bucket name: *myfrontend-appbucket*
- Region: same as your ALB
- **Uncheck** “Block all public access”
- Create the bucket



3. Enable Static website hosting

- Open your bucket
- Go to **Properties**
- Scroll to **Static website hosting**
- Enable it
- Set:
- Index document: *index.html*
- Save changes



4. Upload Files

1. Go to Objects → Upload

2. Upload: Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Static Frontend - Cloud Project</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="container">
    <h1>Static Frontend (S3 + CloudFront)</h1>
    <p>This frontend is hosted on Amazon S3 and served via CloudFront.</p>

    <button onclick="callBackend()">Call Backend API</button>

    <div id="response-box">
      <h3>Backend Response:</h3>
      <p id="api-response">Waiting for response...</p>
    </div>
  </div>

  <script src="app.js"></script>
</body>
</html>
```

App.j

```
function callBackend() {
  // Replace <ALB-DNS> with your actual ALB DNS name
  const backendUrl = "http://taskappjs-1714280081.us-east-1.elb.amazonaws.com/api";

  fetch(backendUrl)
    .then(response => {
      if (!response.ok) {
        throw new Error("Backend not reachable");
      }
      return response.text();
    })
    .then(data => {
      document.getElementById("api-response").innerText = data;
    })
    .catch(error => {
      document.getElementById("api-response").innerText =
        "Error connecting to backend";
      console.error(error);
    });
}
```

Style.css

```
body {
  font-family: Arial, sans-serif;
  background-color: #f4f6f8;
  margin: 0;
  padding: 0;
}

.container {
  width: 60%;
  margin: 80px auto;
  padding: 30px;
  background-color: #ffffff;
  text-align: center;
  border-radius: 8px;
  box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
}

button {
  background-color: #0073e6;
  color: white;
  border: none;
  padding: 12px 25px;
  font-size: 16px;
  cursor: pointer;
  border-radius: 5px;
}

button:hover {
  background-color: #005bb5;
}

#response-box {
  margin-top: 30px;
  padding: 15px;
  border: 1px solid #ddd;
  border-radius: 5px;
}

#api-response {
  color: green;
  font-weight: bold;
}
```

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Storage management and insights

Storage Lens

Batch Operations

Account and organization settings

AWS Marketplace for S3

myfrontend-appbucket

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (3)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	app.js	js	December 22, 2025, 11:26:14 (UTC+05:30)	268.0 B	Standard
☐	index.html	html	December 22, 2025, 11:26:58 (UTC+05:30)	288.0 B	Standard
☐	style.css	css	December 22, 2025, 11:26:38 (UTC+05:30)	46.0 B	Standard

5. Create CloudFront Distribution

CloudFront

Distributions

Create distribution

Step 1

Choose a plan

Step 2

Get started

Step 3

Specify origin

Step 4

Enable security

Step 5

Get TLS certificate

Step 6

Review and create

Choose a plan

Everything you need for a simple monthly price

Plans include global CDN, WAF, DDoS protection, DNS, TLS certificate, log ingestion, and serverless edge compute.

No overage charges.

Blocked requests and DDoS attacks never count against your usage allowance.

Data transfer costs from your AWS origins (such as Amazon S3, Application Load Balancer, or API Gateway) to CloudFront are automatically waived.

Free

\$0/month

For hobbyists, learners, and developers getting started.

Usage allowance

1M requests / 100GB per month

Always-on DDoS protection

Protect against common web threats with AWS WAF

IP-based rate limiting

Geographic traffic blocking

Serverless edge compute

Smart routing

Global CDN

DNS

Free TLS certificate

Tiered caching

Default caching rules

Fast cache invalidations

5GB S3 storage included

Pro

\$15/month

Launch and grow small websites, blogs, and applications.

Business

\$200/month

Protect and accelerate business applications.

Premium

\$1000/month

Scale and protect business and mission-critical applications.

Step 3

Specify origin

Step 4

Enable security

Step 5

Review and create

Origin type

Amazon S3

Deliver static assets like files and images, statically generated websites or single page applications (SPA).

Elastic Load Balancer

Deliver applications hosted behind ELB such as dynamic websites, web services, and APIs.

API Gateway

Deliver API endpoints for REST APIs hosted on API Gateway.

Elemental MediaPackage

Deliver end-to-end live events or video on demand (VOD).

VPC origin

Deliver applications and content hosted within private VPCs, such as EC2 instances and Application Load Balancers.

Other

Refer to any AWS or non-AWS origin through its publicly resolvable URL.

Origin

S3 origin

Choose an AWS origin, or enter your origin's domain name.

myfrontend-appbucket.s3-us-east-1.amazonaws.com

Use website endpoint

This S3 bucket has static web hosting enabled. If you plan to use this distribution as a website, we recommend using the S3 website endpoint rather than the bucket endpoint.

Origin path - optional

The directory path within your origin where your content is stored.

/path

And create distribution


```

GNU nano 8.3
// app.js
const express = require("express");
const app = express();

app.get("/api", (req, res) => {
  res.json({ message: "Backend is working" });
});

app.listen(3000, '0.0.0.0', () => {
  console.log("Server running on port 3000");
});

```

Run to check working or not.

```

{"message":"Backend is working"}[ec2-user@ip-10-0-2-96 ~]$ node app.js
Server running on port 3000
[ec2-user@ip-10-0-2-96 ~]$

```

5. Create an Application Load Balancer (ALB)

- Go to EC2 → Load Balancers → Create Load Balancer → Application Load Balancer.
- Scheme: internet-facing if the frontend needs to access it publicly.
- Listeners: HTTP 80 (or HTTPS 443 if you have a certificate).
- Subnets: choose public subnets.
- Security group: allow inbound HTTP/HTTPS from anywhere or restrict to frontend CloudFront if needed.

EC2 > Load balancers > Create Application Load Balancer

Name must be unique within your AWS account and can't be changed after the load balancer is created.

task 5 hybrid app

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme Info

Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type Info

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**

- Includes only IPv4 addresses.

☐ **Dualstack**

- Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**

- Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

Network mapping Info

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC Info

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-01dab04e5b7d5eb7d (My-Vpc)

10.0.0.0/16

Create VPC

Select subnets and security group for ALB

EC2 > Load balancers > Create Application Load Balancer

IP pools Info
 You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).
☐ Use IPAM pool for public IPv4 addresses
 The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets Info
 Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ us-east-1a (use1-a24)
Subnet
 Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
 subnet-0af133a5591a33c48
 IPv4 subnet CIDR: 10.0.1.0/24 publicsubnet

Security groups Info
 A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).
Security groups
 Select up to 5 security groups
 ALB-SG
 sg-08f59a54633d22ef VPC: vpc-01dab04e5b7d5eb7d

Listeners and routing Info
 A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Select forward to target groups and crate a new target group

EC2 > Load balancers > Create Application Load Balancer

Default action Info
 The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action
☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group Info
 Choose a target group and specify routing weight or [create target group](#).
Target group
 taskappjs HTTP
 Target type: Instance, IPv4 | Target stickiness: Off
 Weight: 1 100%
 0-999
 + Add target group
 You can add up to 4 more target groups.

Target group stickiness Info
 Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.
☐ Turn on target group stickiness

Listener tags - optional
 Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.
 Add listener tag
 You can add up to 50 more tags.

6. Target Group for EC2

- Create a target group → Type: instance → Protocol: HTTP → Port: 3000 (your backend port).
- Register your EC2 instance with this target group.
- Attach the target group to the ALB listener

Select target type - instances

EC2 > Target groups > Create target group

Step 3
Review and create

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name
Name must be unique per Region per AWS account.
taskappjs
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 9/32

Protocol
Protocol for communication between the load balancer and targets.
HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
3000
1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.
☒ IPv4

Select your VPC

the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-01dab04e5b7d5eb7d (My-Vpc)
10.0.0.0/16

[Create VPC](#)

Protocol version

☒ **HTTP1**
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ **HTTP2**
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ **gRPC**
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Register target – private instance

EC2 > Target groups > Create target group

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Register targets - recommended
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (1/2)

Filter instances

	Instance ID	Name	State	Security groups	Zone	Private IP
☐	i-08c810653897946da	PublicBastion	Running	launch-wizard-5	us-east-1a	10.0.1.254
☒	i-012883371274f9db1	Private	Running	EC2-SG	us-east-1a	10.0.2.96

1 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.
3000
1-65535 (separate multiple ports with commas)

[Include as pending below](#)

1 selection is now pending below. Include more or register targets when ready.

Attach this target group to ALB and register our private instance to target.

7. Verify backend access

- test backend directly from a machine with internet access (frontend should call ALB)

curl http://<ALB-DNS>/api

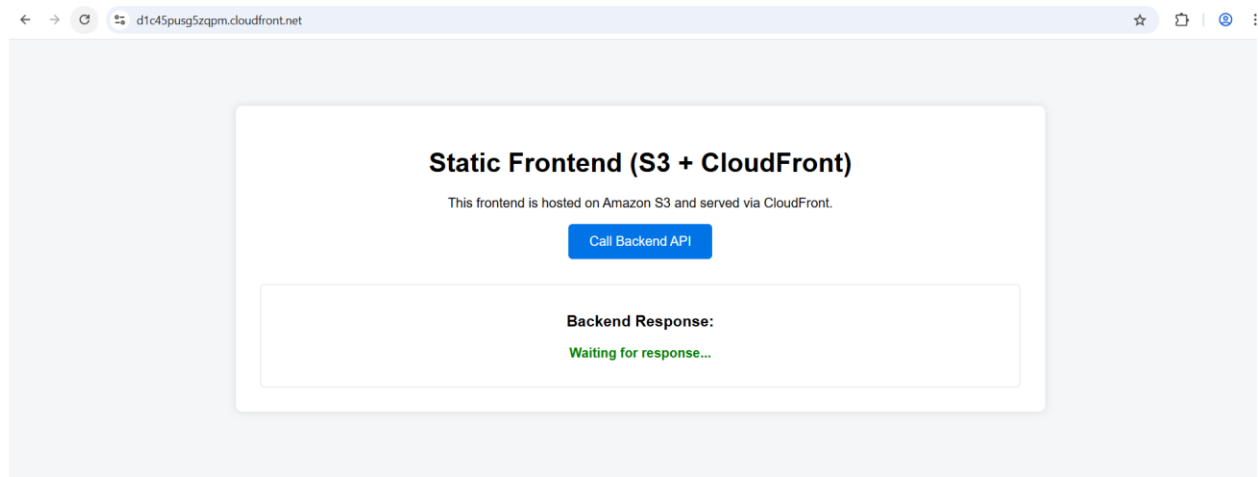
```
[ec2-user@ip-10-0-2-96 ~]$ curl http://taskappjs-1714280081.us-east-1.elb.amazonaws.com/api
{"message":"Backend is working"}[ec2-user@ip-10-0-2-96 ~]$
```

Also check with only ALB-DNS

curl http://<ALB-DNS>

```
[ec2-user@ip-10-0-2-96 ~]$ curl http://localhost:3000
Backend is running[ec2-user@ip-10-0-2-96 ~]$ curl http://taskappjs-1714280081.us-east-1.elb.amazonaws.com
Backend is running[ec2-user@ip-10-0-2-96 ~]$
```

For front end:



Edit outbound rules [Info](#)

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules [Info](#)
Security group rule ID
sgr-0b1369fbf162005ed

Type [Info](#)
All traffic ▼

Protocol [Info](#)
All

Port range [Info](#)
All

Destination [Info](#)
Custom ▼

Q
0.0.0.0/0 X

Description - optional [Info](#)

Delete

Add rule

⚠ Rules with destination of 0.0.0.0/0 or ::0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

CancelPreview changesSave rules